Another interesting thing to note about Tables 8.4, 8.5, and 8.6 is that the numbers level off at a 60% allocation to equity. In other words, any additional allocation doesn't do much to improve the probability of sustainability. Why is this? Shouldn't higher levels of equity continue to increase the sustainability numbers? Well, the fact is that higher equity allocations will increase the growth rate of your portfolio in the long run. But in the short run, there's always some risk of a bad run of luck. Stated differently, with a 100% allocation to equities, there's a 9% to 15% chance that in the first few years you will experience negative returns. The consequent damage to your capital base is too severe to recover in such a short time horizon. To give a sense of the richness of this formula approach, let's take a look at another set of results. This time, let's consider a 55-year-old, perhaps interested in early retirement and withdrawing only 6%.

From Table 8.5, we can see that a 55-year-old who consumes $12,000 every year from an original nest egg of $200,000 (which represents a 6% needs-to-wealth ratio) has only a 29.6% chance of success or a 70.4% chance of outliving her wealth, if all the money is invested in money market funds. Essentially, with that kind of a low-volatility (and low-return) asset allocation, there is little hope for a successful early retirement. At the other extreme, with a 100% allocation to equities, the odds of success are increased. As you can see, the probability of sustainability increases to 64% or alternatively stated, the chances of outliving wealth is reduced to 36% with a complete allocation to the equity market. Granted, a 36% chance of running short is still rather intimidating, and not tolerable by any means. But the significant reduction from 70% to almost 40% is persuasive, and yields insight into the power of long-term equity growth, despite the associated volatility.

Of course, one could explore countless scenarios for various ages and needs-to-wealth strategies. But in the limited set of presented results, the main relationships emerge.

Finally, Figure 8.4 takes a rather bleak perspective on the issue of retirement probabilities and displays the opposite of the sustainability ratio—the probability of retirement ruin. Yes, an ugly word but I believe it helps shed some additional light on the issue at hand.